

Critical Memory in Pricing Learners: Exponential Discovery Gaps in Routing Markets

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Abstract

A marketplace can make a profitable price cut hard to learn without making it unprofitable. We study an inference router that discounts a provider after a cut until its trailing price history is cleared. The resulting pricing MDP has an exact rational memory boundary. Below it, cutting is optimal; above it, staying high is optimal. We prove a separate finite-time boundary: if a learner chooses the exploratory cut with conditional probability at most q , discovering a length- M cut path by time T has probability at most Tq^M . Thus the memory required to defeat a fixed learning horizon grows only logarithmically in T , even when post-discovery rewards have high signal. A path-equivalent commitment option reduces exponential discovery time to linear while preserving every primitive price path and the optimal value. In a frozen calibrated routing game, primitive Q-learning selects the rational action in 1/20 seeds at $M = 7$, versus 19/20 with the option; the paired regret change is -0.0643 (seed-bootstrap 95% interval $[-0.0755, -0.0493]$). The intervention reverses beyond the rational boundary, where primitive learning is correct and commitment overcuts. Full state and an eight-step TD target do not close the gap. The result isolates a temporal credit mechanism; it is not evidence of live-provider collusion.

1 Introduction

Independent pricing algorithms can coordinate through market feedback even without communication. Existing results emphasize reward signal quality, parallel experimentation, and the scoring rule faced by each learner (Hansen et al., 2021; Calvano et al., 2020). A different friction appears when the market itself remembers actions. A price cut can pay poorly today, yet become profitable only after enough consecutive cuts erase a router penalty. No single experiment reveals the long-run return.

AI inference routing provides a concrete instance. Providers sell execution of a common open-weight model, and a router maps posted prices and eligibility signals into request allocation. For a documented price-weighted rule, selection weight is proportional to inverse squared price (OpenRouter, 2026). Public endpoint aggregates also associate recent price cutting with lower selection in one buyer tier. We do not interpret that association causally. We ask what learning problem such a history-dependent

allocation rule would create if imposed.

This distinction matters for both algorithmic-collusion and mechanism-design claims. A high learned price can arise because high pricing is an equilibrium, because experiments are statistically coupled, or because a learner never discovers a profitable delayed path. These mechanisms demand different interventions. More exploration addresses the last mechanism only at an exponential cost; changing the action interface can solve it but may induce overcommitment.

Our contributions are fourfold.

1. We derive an exact *rational* boundary M^* for persistent cutting under a finite-memory penalty. It separates environments where the long-run low price is optimal from those where the high price is optimal.
2. We derive a distinct *learning* boundary. Under any adaptive policy whose pre-discovery cut probability is at most q , $\mathbb{P}(\tau_M \leq T) \leq Tq^M$. Combining this bound with ordinary reward estimation separates temporal discovery from reward SNR.
3. We show that a length- $(M + 1)$ commitment option preserves the optimal value and feasible primitive paths while replacing exponential discovery time by a linear term. This is an interface result, not a change in provider payoffs.
4. We test the mechanism in a frozen, calibrated finite MDP. The option closes the local gap at intermediate memory and becomes harmful beyond the rational boundary. State observability, cross-market severity, and a conventional multi-step return each fail prespecified tests.

2 Routing game and economic boundary

One market is a model-workload pair. Provider i posts price p_i , has marginal cost c_i , and receives first-route share

$$s_i(p) = \frac{p_i^{-a}}{\sum_j p_j^{-a}}, \quad a > 0. \quad (1)$$

With fixed demand, profit is $s_i(p)(p_i - c_i)$. The rule gives $\partial \log s_i / \partial \log p_i = -a(1 - s_i)$; hence the router exponent sets the own-price elasticity directly. At a symmetric interior equilibrium,

$$p^* = c \frac{a(n-1)}{a(n-1) - n}, \quad (2)$$

which exists only if $a(n-1) > n$. The deployed value $a = 2$ is the duopoly knife edge. This static fact motivates the environment but is not our main learning result.

Now fix rivals and restrict one provider to a high quote H and low quote L . A low quote is penalized until the previous M quotes are also low. Write u_H for profit at H , $u_{\theta L}$ for penalized profit at L , and u_L for unpenalized profit at L . The economically interesting ordering is

$$u_L > u_H > u_{\theta L}. \quad (3)$$

Thus cutting loses while the flag is active but wins after it clears.

Theorem 1 (Rational memory boundary). *From an all-high history, an infinite-horizon optimal policy either stays high forever or cuts immediately and stays low. For discount $\gamma \in (0, 1)$, cutting is strictly optimal iff*

$$\gamma^M > \frac{u_H - u_{\theta L}}{u_L - u_{\theta L}}. \quad (4)$$

The continuous boundary is $M^* = \log[(u_H - u_{\theta L})/(u_L - u_{\theta L})]/\log \gamma$.

In the frozen calibrated profile, $(u_H, u_{\theta L}, u_L) = (0.1067535, 0.0351280, 0.1501829)$ and $\gamma = 0.95$, giving $M^* = 9.2402$. Exact dynamic programming cuts for integer $M \leq 9$ and stays high for $M \geq 10$. This boundary is derived before examining learning outcomes.

3 Critical memory for finite-time learning

Let $A_t \in \{H, L\}$ and let $\tau_M = \inf\{t : A_{t-M+1} = \dots = A_t = L\}$ be the first time the learner reaches the unpenalized low state. Before τ_M , rewards from L are below the high reward, so a myopic or optimistic learner can assign it only bounded exploration probability. We allow full history dependence and require only

$$\mathbb{P}(A_t = L \mid \mathcal{F}_{t-1}, \tau_M \geq t) \leq q. \quad (5)$$

Theorem 2 (Exponential discovery barrier). *Under equation (5), for every horizon T ,*

$$\mathbb{P}(\tau_M \leq T) \leq (T - M + 1)_+ q^M \leq T q^M. \quad (6)$$

Consequently, attaining discovery probability at least $1 - \delta$ requires

$$T \geq (1 - \delta) q^{-M}. \quad (7)$$

If pre-discovery actions are iid Bernoulli(q), the exact expected time is

$$\mathbb{E}\tau_M = \frac{1 - q^M}{(1 - q)q^M}. \quad (8)$$

Define the finite-horizon critical memory

$$M_c(T, q, \delta) = \left\lceil \frac{\log[T/(1 - \delta)]}{\log(1/q)} \right\rceil. \quad (9)$$

For $M \geq M_c + 1$, the target success probability is ruled out regardless of how accurately the learner estimates rewards after discovery. Thus memory is a computational control parameter: increasing T by a factor of ten buys only $1/\log_{10}(1/q)$ additional memory periods.

This clarifies the relationship to signal-to-noise results. Suppose post-discovery reward differences are sub-Gaussian with gap Δ and proxy variance σ^2 . After discovery, $O(\text{SNR}^{-1} \log(1/\delta))$ samples, where $\text{SNR} = \Delta^2/\sigma^2$, suffice to identify the better stationary action. Before discovery, equation (7) still applies.

Corollary 3 (Temporal–statistical crossover). *For a fixed-confidence learner, total difficulty contains a temporal term q^{-M} and a statistical term $\text{SNR}^{-1} \log(1/\delta)$. Temporal discovery dominates once*

$$q^{-M} \gg \text{SNR}^{-1} \log(1/\delta). \quad (10)$$

Higher reward SNR cannot remove this gap; it only improves learning after the rare state has been reached.

The result complements Hansen–Misra–Pai: independent experiments may couple through the final pricing score even with no explicit memory (Hansen et al., 2021). Our theorem does not contradict asymptotic learnability. It supplies a finite-time rate: once market memory requires a run rather than an isolated experiment, the time needed to expose the relevant score can be exponential in M .

4 A path-equivalent intervention

Augment the primitive MDP with a `commit-low` option that executes $M + 1$ existing low actions. Both primitive actions remain available.

Theorem 4 (Value preservation and linear discovery). *The augmented semi-Markov MDP has exactly the same optimal value as the primitive MDP at every state. It creates no new primitive price path. If the option is selected independently with probability q_o at each decision epoch while non-option decisions consume one transition, its completion time has expectation*

$$M + q_o^{-1}. \quad (11)$$

The option changes representation, not incentives. Any augmented policy can be unrolled into the same sequence of primitive actions and discounted rewards. But the probability of proposing the full low path is now q_o rather than q^M . This is the exponential temporal-abstraction gap shown in figure 1A.

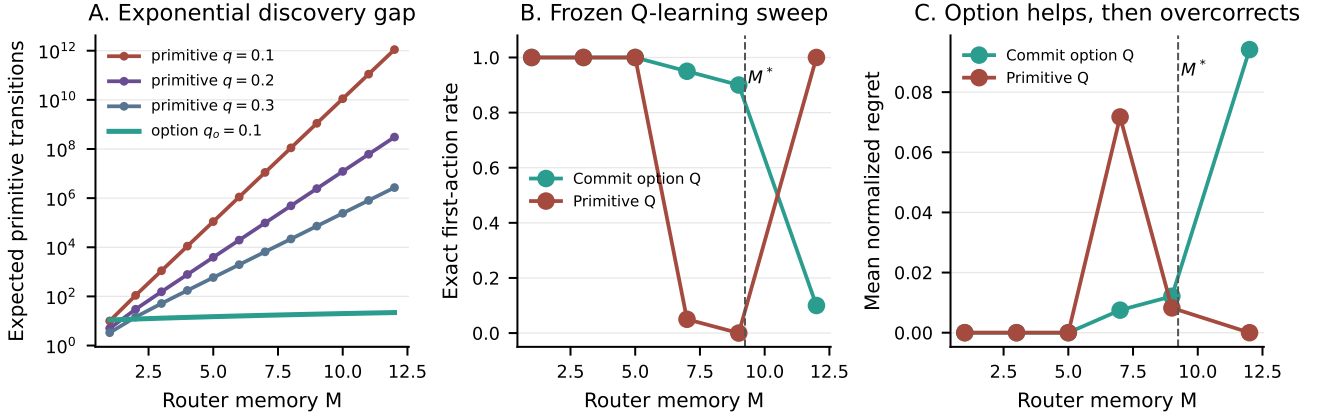


Figure 1: Theory and frozen experiment. A: iid primitive exploration takes exponential time to discover M consecutive cuts; a commitment option is linear in M . B: exact first-action agreement over 20 seeds. C: mean normalized regret. The dashed rational boundary $M^* = 9.2402$ is computed from payoffs, not fitted to learner outcomes.

5 Experimental design

The experiment is a controlled finite-MDP mechanism test, not a reduced-form test on provider prices. Its high and low quotes and router penalty are frozen from an audited market profile. Exact value iteration supplies the target action. Tabular Q-learning uses 300,000 primitive transitions, discount 0.95, and 20 prespecified seeds. The option learner receives the same primitive transition budget; an option consumes its full duration. A 10,000-transition frozen-policy evaluation reports first-action agreement and normalized regret. Paired seed bootstrap intervals describe simulation variation only.

We use a falsification ladder rather than interpreting one favorable run. E-SIM5 gives the primitive learner the complete Markov history. E-SIM6 adds only the path-equivalent option and sweeps $M \in \{1, 3, 5, 7, 9, 12\}$. E-SIM7 transports the intervention to four frozen price books. E-SIM8 varies learning rate and exploration decay in a 3×3 grid. E-SIM9 replaces one-step Q with an eight-step TD target while retaining the primitive action set. All gates, seeds, and result hashes were frozen before interpretation.

The live empirical bridge is deliberately weaker. In the latest hosted panel, the primary same-regime covariance statistic exceeds its randomization null by 0.0256 with one-sided $p = 0.0500$, but one provider-pair accounts for 87.3% of eligible events, the identity-shuffle diagnostic has $p = 0.992$, and downstream family gates are underpowered. This supports studying signal-coupling mechanisms; it does not validate them as provider conduct.

6 Results

Critical memory is visible in learning. At $M \in \{1, 3, 5\}$, primitive and option learners agree with exact control in 20/20 seeds. At $M = 7$, where exact opti-

mization still cuts, primitive Q agrees in 1/20 seeds while option Q agrees in 19/20; 18/20 option seeds also satisfy the joint low-regret gate. At $M = 9$, the comparison is 0/20 versus 18/20. The transition is therefore near, but below, the independently derived rational boundary.

At $M = 7$, mean normalized regret is 0.07175 for primitive Q and 0.00750 for option Q. The paired difference is -0.06425 with 95% seed-bootstrap interval $[-0.07553, -0.04930]$. The sign remains negative in every one of nine learning-rate/exploration-decay cells; the strict composite severity gate passes in seven.

The intervention is nonmonotone. At $M = 12$, exact optimization stays high. Primitive Q is correct in 20/20 seeds, whereas option Q is correct in only 2/20 and adds 0.0942 normalized regret. An interface that solves delayed discovery can therefore reduce welfare when it creates commitment salience on the wrong side of the economic boundary. Mechanism evaluation must combine theorem 1 with theorem 2; the learning boundary alone is not a design objective.

The falsifications narrow the mechanism. Complete state observability does not repair primitive learning: it remains high in 19/20 seeds at $M = 7$. The severe-trap transport gate fails in two of four external price books, although all four option-effect signs align with their rational boundary. Finally, an ordinary eight-step TD target succeeds in 0/20 seeds and has regret difference $+0.0038$ relative to one-step Q, interval $[0, 0.0113]$. The supported result is option-specific temporal abstraction in the controlled MDP, not a claim that any longer return solves delayed credit.

7 Implications for pricing mechanisms

The two boundaries separate three regimes.

1. If $M > M^*$, a persistent cut destroys provider value. High pricing is rational; exploration policy is secondary.
2. If $M \leq M^*$ but $M > M_c(T, q, \delta)$, cutting is rational but is unlikely to be discovered within the learning horizon. A platform can sustain high learned prices without communication or supra-rational reward coupling.
3. If $M \leq \min\{M^*, M_c\}$, ordinary exploration can expose the low path; reward SNR and cross-agent signal coupling again become first-order.

This yields a design warning. Router memory can look anti-gaming because it disciplines short-lived cuts, yet the same rule can suppress efficient persistent cuts by altering learnability. Conversely, exposing a commitment action can restore price competition below M^* but cause excessive cutting above it. A safer router would expose the state and a cancellable multi-period quote schedule only when a deviation audit predicts that persistent cutting raises provider value and buyer surplus. The theorem supplies a testable screen, not a universally optimal policy.

8 Related work

Algorithmic-pricing research establishes that independent learners may support high prices and that results depend sharply on algorithms and timing (Calvano et al., 2020; Asker et al., 2022; Brown and MacKay, 2023). Hansen–Misra–Pai show how independent bandit experiments become strategically coupled through the aggregate price used to score arms (Hansen et al., 2021). Johnson–Rhodes–Wildenbeest study platform steering when sellers use pricing algorithms (Johnson et al., 2023). Our contribution is orthogonal: a router’s own-history rule creates a finite-time run-discovery barrier even when reward differences are otherwise easy to estimate.

Options and semi-Markov value preservation are classical (Sutton et al., 1999); delayed reward and sequence compression have recently been studied directly (Han et al., 2022; Ramesh et al., 2024). We add an economic source of the delay, an explicit critical-memory rate, a rational boundary with the opposite comparative-static force, and controlled evidence that the representation intervention helps only between those boundaries.

9 Limitations and claim boundary

The finite-time theorem assumes a pre-discovery cut-probability cap; it does not characterize algorithms that plan over a known model or deliberately schedule low runs. The calibrated experiment fixes rivals, uses two prices, and studies tabular Q. Twenty seeds identify Monte

Carlo variability, not uncertainty in live-market parameters. Cross-market severe-trap transport fails, and no paid probe randomizes router memory. The empirical covariance screen is concentrated and family-incomplete. We therefore claim a theorem, a controlled mechanism, and a reproducible local counterfactual—not provider intent, tacit collusion, communication, or live-router causality.

10 Conclusion

Market memory creates two distinct thresholds. The rational threshold says whether persistent cutting is valuable; the finite-time threshold says whether a bounded learner can discover it. Between them, high learned prices can persist despite a profitable low-price path, with discovery time exponential in memory and only logarithmic improvement from longer training. A path-equivalent option collapses that gap but overcorrects when cutting is no longer rational. Learning-aware marketplace design must therefore optimize both incentives and representations, then audit deviations on either side of the boundary.

A Proofs

Proof of theorem 1. Once the history is all-low, L strictly dominates H by equation (3); choosing H also restarts the penalty. Before that state, any policy that eventually cuts is weakly improved to k high actions followed by low forever. Its value is

$$V_k = \frac{u_H(1 - \gamma^k)}{1 - \gamma} + \gamma^k V_{\text{cut}},$$

$$V_{\text{cut}} = \frac{(1 - \gamma^M)u_{\theta L} + \gamma^M u_L}{1 - \gamma}.$$

Because $V_k - u_H/(1 - \gamma) = \gamma^k[V_{\text{cut}} - u_H/(1 - \gamma)]$, either $k = 0$ or never cutting is optimal. Comparing these alternatives yields equation (4). $\square$

Proof of theorem 2. For each candidate ending time t , iterated conditioning and equation (5) bound the probability of the required all-low length- M window by q^M . A union bound over $(T - M + 1)_+$ windows proves equation (6); inversion of its looser Tq^M form gives equation (7). Under iid Bernoulli exploration, let E_k be the expected remaining time after a current run of k lows. The recursions $E_M = 0$ and $E_k = 1 + qE_{k+1} + (1 - q)E_0$ solve to $E_0 = (1 - q^M)/[(1 - q)q^M]$. $\square$

Proof of theorem 4. The augmented action set contains every primitive action, so its optimal value is weakly larger. Replace each option in any augmented policy by its defining $M + 1$ primitive low actions. The unrolled policy produces identical states, durations, and discounted rewards, so its value is identical and the augmented optimum cannot be larger. Before the first option, the number of one-step decisions is geometric with mean $(1 - q_o)/q_o$; adding the final $M + 1$ transitions gives $M + q_o^{-1}$. $\square$

B Reproducibility

The theory functions and unit tests are in `src/orcap/market_env/critical_memory.py` and `tests/market_env/test_critical_memory.py`. The figure is generated directly from the frozen E-SIM6 parquet. Clean bundles are E-SIM5 `3e5a55405e`, E-SIM6 `93265b9b03`, E-SIM7 `bd74ab2eb7`, E-SIM8 `4d84b9b3a2`, and E-SIM9 `179eca0f9d`. Manifests pin source commits, seeds, result hashes, and input-artifact hashes. Exact Bellman residuals, path equivalence, paper numbers, and the new finite-time identities are enforced by tests.

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